

Quantum Information Theory

Lesson 4: Quantum Random Variables

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Postulates of Quantum Mechanics

1. The momentary state of an isolated system is a vector $|\psi(t)\rangle \in \mathcal{H}(S)$ in a complex Hilbert space $\mathcal{H}(S)$;
2. Measurable physical quantities are described by Hermitian operators (observables, $\mathbf{O} = \sum_i \lambda_i |\psi_i\rangle \langle \psi_i|$);
3. Measuring an observable produces one of its eigenvalues;
4. Probability of measuring λ is $\sum_{\lambda_i=\lambda} \langle \psi_i | \psi(t) \rangle \langle \psi(t) | \psi_i \rangle$;
5. When λ is observed, the post-measurement state is

$$|\psi_\lambda(t+1)\rangle \propto \sum_{\lambda_i=\lambda} |\psi_i\rangle \langle \psi_i | \psi(t) \rangle ;$$

6. The time evolution of $|\psi(t)\rangle$ is governed by Schrödinger's equation, $i\hbar \frac{d}{dt} |\psi(t)\rangle = \mathbf{H}(t) |\psi(t)\rangle$, where $H(t)$ is the observable associated with the total energy (Hamiltonian).

Quantum Horses

We are simulating virtual races of 6 horses with win probabilities

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}.$$

We want to generate the random results using a quantum system.

Then, W must correspond to a quantum observable $\mathbf{W}$.

This $\mathbf{W}$ needs to have a finite eigenvalue λ for each outcome.

So, it can be a 6×6 Hermitian matrix $\mathbf{W} \in \mathcal{B}(\mathbb{C}^6) \subseteq \text{Mat}_6(\mathbb{C})$.

Given that we need to observe this 6×6 $\mathbf{W}$, we need $|\psi\rangle \in \mathbb{C}^6$.

We don't *need* to use 6×6 , but that is the minimal choice.

If we were to use a 8×8 matrix, some λ s need to be repeated.

This observable corresponds to 8-lines vectors $|\psi\rangle \in \mathbb{C}^8 \cong \mathbb{C}^{2^{\otimes 3}}$.

For theory, we use the minimal choice, as the formulas are simpler.

For implementation, if all we have is qubits we will round to 2^N .

We may even consider packing e.g. $\frac{6^5}{2^{\lceil \log_2 6^5 \rceil}} \approx 95\% \gg 75\% = \frac{6}{8}$.

Quantum Horses

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We want to generate the random results using a quantum system.

We need the state-space to have at least 6 dimensions.

This would be a d -level quantum system (**qudit**) with $d = 6$.

We will sample this W using a quantum observable $\mathbf{W}$.

But the outcome of the measurement is one of the eigenvalues.

And these eigenvalues must be real numbers.

So we are forced to identify A, B, C, D, E, F with numbers.

In itself, this is completely legitimate, but it may be misleading.

For example, the average of the measured result is meaningless.

The variance used in e.g. Heisenberg's principle is also meaningless.

Change your choice of eigenvalues and these statistics will change.

Quantum Horses

We are simulating virtual races of 6 horses with win probabilities

$$\mathbf{W} \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}.$$

We want to generate the random results using a quantum system, by measuring an observable $\mathbf{W}$ upon a quantum state $|\psi\rangle \in \mathbb{C}^6$.

Let's say $\mathbf{W}$ has spectral decomposition $\mathbf{W} = \sum_{k=1}^6 \lambda_k |\phi_k\rangle \langle \phi_k|$.
Let's say we measure $\mathbf{W}$ upon $|\psi\rangle = \sum_{i=1}^6 \alpha_i |i\rangle$. We will get

$$\mathbf{W} \sim \begin{pmatrix} \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & \lambda_6 \\ |\langle \psi | \phi_1 \rangle|^2 & |\langle \psi | \phi_2 \rangle|^2 & |\langle \psi | \phi_3 \rangle|^2 & |\langle \psi | \phi_4 \rangle|^2 & |\langle \psi | \phi_5 \rangle|^2 & |\langle \psi | \phi_6 \rangle|^2 \end{pmatrix},$$

so if we identify $A \leftrightarrow \lambda_1$, $B \leftrightarrow \lambda_2$, $\dots$, $F \leftrightarrow \lambda_6$ we want to have $|\langle \psi | \phi_1 \rangle|^2 = 0.1$, $|\langle \psi | \phi_2 \rangle|^2 = 0.2$, $\dots$, $|\langle \psi | \phi_6 \rangle|^2 = 0.2$.

Notice that $|\psi\rangle$ has 6 complex coefficients (11 independent reals). But we only have 6 constraints. There are infinite choices for $|\psi\rangle$.

Quantum Horses

We are simulating virtual races of 6 horses with win probabilities

$$\mathbf{W} \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}.$$

We want to generate the random results using a quantum system, by measuring an observable $\mathbf{W}$ upon a quantum state $|\psi\rangle \in \mathbb{C}^6$. Let's say $\mathbf{W}$ has spectral decomposition $\mathbf{W} = \sum_{k=1}^6 \lambda_k |\phi_k\rangle \langle \phi_k|$.

We will express $|\psi\rangle$ in the $\{|\phi_k\rangle\}$ basis, not the computational one. If $|\psi\rangle = \sum_i \alpha_i |\phi_i\rangle$ we get $|\langle \psi | \phi_k \rangle|^2 = |\alpha_k|^2$ so we can take e.g.

$$|\psi\rangle = \sqrt{0.1} |\phi_1\rangle + \sqrt{0.2} |\phi_2\rangle + \sqrt{0.05} |\phi_3\rangle + \sqrt{0.15} |\phi_4\rangle + \sqrt{0.3} |\phi_5\rangle + \sqrt{0.2} |\phi_6\rangle$$

$$|\psi\rangle = \sqrt{0.1} |\phi_1\rangle + \sqrt{0.1}(1+i) |\phi_2\rangle + \sqrt{0.05} |\phi_3\rangle + \sqrt{0.15} |\phi_4\rangle + \sqrt{0.3} |\phi_5\rangle + \sqrt{0.2} |\phi_6\rangle$$

$$|\psi\rangle = \sqrt{0.1} |\phi_1\rangle + \sqrt{0.1}(1+i) |\phi_2\rangle + \sqrt{0.01}(2+i) |\phi_3\rangle + \sqrt{0.15} |\phi_4\rangle + \sqrt{0.3} |\phi_5\rangle + \sqrt{0.2} |\phi_6\rangle$$

$$|\psi\rangle = \sqrt{0.1} |\phi_1\rangle + \sqrt{0.225} |\xi_2\rangle + \sqrt{0.025} |\xi_3\rangle + \sqrt{0.15} |\phi_4\rangle + \sqrt{0.3} |\phi_5\rangle + \sqrt{0.2} |\phi_6\rangle,$$

$$\text{where } |\xi_2\rangle = \frac{|\phi_2\rangle + |\phi_3\rangle}{\sqrt{2}}, |\xi_3\rangle = \frac{|\phi_2\rangle - |\phi_3\rangle}{\sqrt{2}}.$$

Indiscernibles

We saw that for any observable $\mathbf{W}$ and desired measurement outcome distribution W there is an infinite number of quantum states $|\psi\rangle$ that can help us achieve our goal.

Since the distributions are the same, if all we do is measure a $\mathbf{W}$ we cannot empirically tell which state $|\psi\rangle$ was measured.

Now, let's say we are given an unknown state $|\nu\rangle$ that is one of two options: $|\psi\rangle$ or $|\xi\rangle$. We ask if there is a $\mathbf{W}$ that lets us tell which.

One recipe to construct such a $\mathbf{W}$ is to extend $|\psi\rangle$ up to a basis. If $|\nu\rangle$ was $|\psi\rangle$, it will collapse into $|\psi\rangle$ with probability 1.

If $|\nu\rangle$ was $|\xi\rangle$, it will collapse into $|\psi\rangle$ with probability $|\langle\xi|\psi\rangle|^2$.

So, if $|\langle\xi|\psi\rangle|^2 \neq 1$ the statistics of this one $\mathbf{W}$ are enough to tell.

On the other hand, if $|\langle\xi|\psi\rangle|^2 = 1$ we have $|\xi\rangle = e^{i\theta}|\psi\rangle$.

So, whatever eigenvectors $\{|\phi_k\rangle\}$ we choose for $\mathbf{W}$, we get the same statistics because $|\langle\xi|\phi_k\rangle|^2 = |\langle\psi|e^{-i\theta}|\phi_k\rangle|^2 = |\langle\psi|\phi_k\rangle|^2$.

Indiscernibles

Two quantum states are called indiscernible if there is no observable whose measurement statistics discriminate them,

$$|\psi\rangle \equiv |\xi\rangle \Leftrightarrow |\psi\rangle = e^{i\theta} |\xi\rangle \Leftrightarrow |\psi\rangle \langle\psi| = |\xi\rangle \langle\xi|.$$

But what about indistinguishable random states? Let's take

$$|\mathbf{v}\rangle \sim \begin{pmatrix} |\psi_1\rangle & |\psi_2\rangle & \dots & |\psi_n\rangle \\ p_1 & p_2 & \dots & p_n \end{pmatrix},$$

and note that the probability that $|\mathbf{v}\rangle$ collapses into $|\phi_k\rangle$ is

$$\mathbb{E} \left[|\langle\psi_i | \phi_k\rangle|^2 \right] = \sum_i p_i |\langle\psi_i | \phi_k\rangle|^2 = \text{Tr} \left(\left(\sum_i p_i |\psi_i\rangle \langle\psi_i| \right) |\phi_k\rangle \langle\phi_k| \right).$$

so $|\mathbf{v}\rangle, |\mathbf{v}'\rangle$ are indistinguishable iff they have the same matrix

$$|\mathbf{v}\rangle \equiv |\mathbf{v}'\rangle \Leftrightarrow \sum_i p_i |\psi_i\rangle \langle\psi_i| = \sum_i p'_i |\psi'_i\rangle \langle\psi'_i|.$$

Mixed States

A random quantum state $|\mathbf{v}\rangle$ has the matrix representation

$$\rho_{|\mathbf{v}\rangle} = \sum_{|\psi\rangle} \text{Pr}[\mathbf{v} = |\psi\rangle] |\psi\rangle \langle\psi|.$$

Looking at these projectors we see that $|\psi\rangle \langle\psi|^\dagger = |\psi\rangle \langle\psi|$, $\text{Tr} |\psi\rangle \langle\psi| = \text{Tr} \langle\psi| |\psi\rangle = 1$ and $\sigma(|\psi\rangle \langle\psi|) = \{0, 1\}$ so $|\psi\rangle \langle\psi| \geq 0$. We conclude that these matrix representations are Hermitian, positive semi-definite and unit-traced, i.e. a density matrix:

$$\begin{aligned}\rho_{|\mathbf{v}\rangle}^\dagger &= \sum_{|\psi\rangle} \text{Pr}[\mathbf{v} = |\psi\rangle] |\psi\rangle \langle\psi|^\dagger = \rho_{|\mathbf{v}\rangle}, \\ \langle\phi_k| \rho_{|\mathbf{v}\rangle} |\phi_k\rangle &= \sum_{|\psi\rangle} \text{Pr}[\mathbf{v} = |\psi\rangle] |\langle\psi | \phi_k\rangle|^2 \geq 0; \\ \text{Tr} \rho_{|\mathbf{v}\rangle} &= \sum_{|\psi\rangle} \text{Pr}[\mathbf{v} = |\psi\rangle] \text{Tr} |\psi\rangle \langle\psi| = 1.\end{aligned}$$

Mixed States

A matrix $\rho \in \mathcal{B}(\mathbb{C}^d)$ is called a density matrix if it is Hermitian, positive semi-definite and unit-traced.

Let's say its spectral decomposition is $\rho = \sum_{i=1}^d \lambda_i |\psi_i\rangle \langle \psi_i|$.

Since $\rho \geq 0$ its eigenvalues are non-negative i.e. $\lambda_i \geq 0 \forall i$.

Since $\text{Tr } \rho = 1$ its eigenvalues sum to 1 i.e. $\sum_{i=1}^d \lambda_i = 1$.

So any density matrix ρ is realized by at least one random variable:

$$|\psi_i\rangle \sim \begin{pmatrix} |\psi_1\rangle & |\psi_2\rangle & \dots & |\psi_d\rangle \\ \lambda_1 & \lambda_2 & \dots & \lambda_d \end{pmatrix},$$

and possibly more e.g. $\frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1| = \frac{1}{2} |+\rangle \langle +| + \frac{1}{2} |-\rangle \langle -|$.

At this point we emphasize that:

- ▶ **A (pure) quantum state $|\psi\rangle \in \mathbb{C}^d$ is a d -rows vector;**
- ▶ **A random quantum state $|v\rangle$ is a function $|v\rangle : \Omega \rightarrow \mathbb{C}^d$;**
- ▶ **A density matrix $\rho \in \mathcal{B}(\mathbb{C}^d)$ is a $d \times d$ matrix.**

Mixed States

Let's say we want to prepare a qudit in state $|\psi\rangle$.

But our equipment is not perfect, so we may have (random) errors.
Our algorithms, protocols etc will have to deal with mixed states.

Let's say we want the perfect superposition e.g. $|\psi\rangle = |+\rangle$.

Maybe the gates are not perfect so what we actually get is e.g.

$$|\psi\rangle_{X,Y} = \sqrt{1-X} |+\rangle + \sqrt{X} e^{2\pi i Y} |-\rangle,$$

with X, Y uniformly distributed over $[0, \epsilon]$ and $[0, 1]$ respectively i.e.

$$\begin{aligned}\rho &= \mathbb{E} [|\psi_{X,Y}\rangle \langle \psi_{X,Y}|] = \mathbf{H} \mathbb{E} \left[\begin{bmatrix} 1-X & \sqrt{X(1-X)} e^{-2\pi i Y} \\ \sqrt{X(1-X)} e^{2\pi i Y} & X \end{bmatrix} \right] \mathbf{H} \\ &= \mathbf{H} \begin{bmatrix} 1 - \frac{\epsilon}{2} & 0 \\ 0 & \frac{\epsilon}{2} \end{bmatrix} \mathbf{H} = \left(1 - \frac{\epsilon}{2}\right) |+\rangle \langle +| + \frac{\epsilon}{2} |-\rangle \langle -| = \Delta_{\epsilon}(|+\rangle \langle +|),\end{aligned}$$

where $\Delta_{\epsilon}(\rho) = (1 - \epsilon)\rho + \epsilon \frac{\mathbf{I}_d}{d}$ is called the depolarizing channel.

Measurement Uncertainty

Of course, if we don't have $|\psi\rangle$ but $\rho = (1 - \epsilon) |\psi\rangle \langle\psi| + \epsilon \frac{\mathbf{I}_d}{d}$, if we measure in a basis including $|\psi\rangle$, it won't always collapse into $|\psi\rangle$.

Considering the observable $\mathbf{O} = \lambda_1 |\psi\rangle \langle\psi| + \sum_{k=2}^d \lambda_k |\phi_k\rangle \langle\phi_k|$,

$$\text{Tr}(\rho |\psi\rangle \langle\psi|) = \text{Tr}\left(\left(1 - \epsilon + \frac{\epsilon}{d}\right) |\psi\rangle \langle\psi| + \mathbf{O}\right) = 1 - \epsilon + \frac{\epsilon}{d},$$

$$\text{Tr}(\rho |\phi_k\rangle \langle\phi_k|) = \text{Tr}\left(\mathbf{O} + \frac{\epsilon}{d} |\phi_k\rangle \langle\phi_k|\right) = \frac{\epsilon}{d},$$

so the measurement results are distributed as

$$\mathbf{O} \sim \begin{pmatrix} \lambda_1 & \lambda_2 & \lambda_3 & \dots & \lambda_d \\ 1 - \frac{d-1}{d}\epsilon & \frac{\epsilon}{d} & \frac{\epsilon}{d} & \dots & \frac{\epsilon}{d} \end{pmatrix},$$

and we can compute e.g. the Shannon entropy

$$\begin{aligned} H(\mathbf{O}) &= - \left(1 - \frac{d-1}{d}\epsilon\right) \log \left(1 - \frac{d-1}{d}\epsilon\right) - \frac{d-1}{d}\epsilon \log \frac{\epsilon}{d} \\ &= h\left(\frac{d-1}{d}\epsilon\right) + \frac{d-1}{d}\epsilon \log(d-1), \end{aligned}$$

and notice that it increases with error parameter ϵ .

Measurement Uncertainty

We can also compute other statistics like the average and variance

$$\langle \mathbf{O} \rangle_\rho := \mathbb{E}[O] = \left(1 - \frac{d-1}{d}\epsilon\right) \lambda_1 + \frac{\epsilon}{d} \sum_{k=2}^d \lambda_k,$$

$$\Delta \mathbf{O}_\rho^2 := \text{var}[O] = \left(1 - \frac{d-1}{d}\epsilon\right) \lambda_1^2 + \frac{\epsilon}{d} \sum_{k=2}^d \lambda_k^2 - \langle \mathbf{O} \rangle_\rho^2,$$

but these measurements depend on the choice of eigenvalues.
For example, if we take $d = 3$, $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$ we get

$$\begin{aligned} \Delta \mathbf{O}_\rho^2 &= \left(1 - \frac{2\epsilon}{3}\right) \cdot 1 + \frac{\epsilon}{3} \cdot 13 - \left(\left(1 - \frac{2\epsilon}{3}\right) \cdot 1 + \frac{\epsilon}{3} \cdot 5\right)^2 \\ &= 1 + \frac{11\epsilon}{3} - (1 + \epsilon)^2 = \frac{5\epsilon}{3} - \epsilon^2 = (5/6)^2 - (\epsilon - 5/6)^2 \end{aligned}$$

which increases up until $\epsilon = 5/6$ then decreases.

Bad monotony is one of the artifacts of badly chosen eigenvalues.

Measurement Uncertainty

In general, for a quantum state $\rho = \sum_{i=1}^d p_i |\psi_i\rangle \langle \psi_i| \in \mathcal{B}(\mathbb{C}^d)$, and observable $\mathbf{O} = \sum_{k=1}^d \lambda_k |\phi_k\rangle \langle \phi_k|$, with possibly equal λ s:

$$\Pr[\mathbf{O} = \lambda] = \sum_{\lambda_k = \lambda} \text{Tr}(\rho |\phi_k\rangle \langle \phi_k|) = \sum_{\lambda_k = \lambda} \langle \phi_k | \rho | \phi_k \rangle,$$

so we have the average, variance and entropy

$$\langle \mathbf{O} \rangle_\rho = \sum_{\lambda} \lambda \Pr[\mathbf{O} = \lambda] = \sum_{k=1}^d \lambda_k \langle \phi_k | \rho | \phi_k \rangle$$

$$\Delta \mathbf{O}_\rho^2 = \langle \mathbf{O}^2 \rangle_\rho - \langle \mathbf{O} \rangle_\rho^2 = \sum_{k=1}^d \lambda_k^2 \langle \phi_k | \rho | \phi_k \rangle - \left(\sum_{k=1}^d \lambda_k \langle \phi_k | \rho | \phi_k \rangle \right)^2$$

$$H(\mathbf{O}) = - \sum_{\lambda} \Pr[\mathbf{O} = \lambda] \log \Pr[\mathbf{O} = \lambda],$$

among which, entropy is our chosen measure of uncertainty.

Measurement Entropy

Looking closer at the formula for measurement entropy, we see that the sum therein is indexed by the unique eigenvalues,

$$H(\mathbf{O}) = - \sum_{\lambda} \Pr[\mathbf{O} = \lambda] \log \Pr[\mathbf{O} = \lambda].$$

For example, for $\mathbf{O} = |00\rangle\langle 00| - |01\rangle\langle 01| - |10\rangle\langle 10| + |11\rangle\langle 11|$ we have 2 λ s so after measuring we may still have superpositions. If we take $\rho = \frac{1}{3} |++\rangle\langle ++| + \frac{2}{3} |00\rangle\langle 00|$ we have

$$\begin{aligned} \langle 00|\rho|00\rangle &= \frac{3}{4}; & \langle 01|\rho|01\rangle &= \frac{1}{12}; & \langle 10|\rho|10\rangle &= \frac{1}{12}; & \langle 11|\rho|11\rangle &= \frac{1}{12}; \\ \Pr[\mathbf{O} = 1] &= \frac{5}{6}; & \Pr[\mathbf{O} = -1] &= \frac{1}{6}; & H(\mathbf{O}) &= \log 6 - \frac{5 \log 5}{6} \approx 0.6500; \end{aligned}$$

and e.g. after observing $\mathbf{O} = -1$ the state collapses to $|\beta_{01}\rangle\langle\beta_{01}|$, and otherwise it collapses into $\rho' = \frac{1}{5} |\beta_{00}\rangle\langle\beta_{00}| + \frac{4}{5} |00\rangle\langle 00|$, which is mixed, so further measurements still produce information.

Measurement Entropy

If we now measure $\mathbf{O}' = |00\rangle\langle 00| - |01\rangle\langle 01| + |10\rangle\langle 10| - |11\rangle\langle 11|$ upon $\rho' = \frac{1}{5} |\beta_{00}\rangle\langle \beta_{00}| + \frac{4}{5} |00\rangle\langle 00|$, we have

$$\begin{aligned}\langle 00|\rho'|00\rangle &= \frac{9}{10}; & \langle 01|\rho'|01\rangle &= 0; & \langle 10|\rho'|10\rangle &= 0; & \langle 11|\rho'|11\rangle &= \frac{1}{10}; \\ \Pr[O' = 1] &= \frac{9}{10}; & \Pr[O' = -1] &= \frac{1}{10}; & H(O') &= \log 10 - \frac{9 \log 9}{10} \approx 0.4690; \\ && \text{and if we measured it upon } |\beta_{01}\rangle && \text{we would get } H(O') &= 1.\end{aligned}$$

In total we have

$$H(O, O') = H(O) + H(O' | O) \approx 0.6500 + \frac{1}{6} + \frac{5}{6} 0.4690 \approx 1.2075$$

which is, notably, the same as if we were to measure both parity bits at once, because the observables commute.

If we take $\mathbf{O}'' = |00\rangle\langle 00| + 2|01\rangle\langle 01| + 3|10\rangle\langle 10| + 4|11\rangle\langle 11|$,

$$H(O'') = -\frac{3}{4} \log \frac{3}{4} - \frac{3}{12} \log \frac{1}{12} \approx 1.2075.$$

Measurement Entropy

We saw that measuring an observable with duplicate eigenvalues does not fully disambiguate the post-measurement state. Such observables correspond to partial measurements.

Basis measurements, also known as von Neumann measurements, correspond to observables with distinct eigenvalues.

If $\mathbf{O} = \sum_{k=1}^d \lambda_k |\phi_k\rangle \langle \phi_k|$ with distinct $\{\lambda_k\}$ we have

$$H(\mathbf{O}) = - \sum_{k=1}^d \langle \phi_k | \rho | \phi_k \rangle \log_2 \langle \phi_k | \rho | \phi_k \rangle ,$$

which is independent of the choice of eigenvalues.

However, this formula still depends on the choice of basis.

We would like to know if there is a “best” or “worst” basis.

von Neumann Entropy

The “worst” basis is the Fourier basis, $|\phi_k\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^d e^{\frac{2\pi i k}{d}} |\psi_i\rangle$, for which we notice that $|\langle\phi_k | \psi_i\rangle|^2 = 1/d$ so $H(\mathbf{O}) = \log d$.

We notice that the vector with entries

$$v_k = \langle\phi_k | \rho | \phi_k\rangle = \langle\phi_k | \sum_{i=1}^d p_i |\psi_i\rangle \langle\psi_i| | \phi_k\rangle = \sum_{i=1}^d \underbrace{|\langle\phi_k | \psi_i\rangle|^2}_{d_{k,i}} p_i$$

can be written as $\mathbf{v} = \mathbf{D}\mathbf{p}$ where the matrix $\mathbf{D}$ is doubly-stochastic so $\mathbf{p} \succeq \mathbf{v}$ so $H(\mathbf{O}) = H(\mathbf{v}) \geq H(\mathbf{p}) = -\sum_{i=1}^d p_i \log p_i$, a quantity known as the von Neumann entropy of ρ ,

$$H(\rho) = -\text{Tr } \rho \log \rho = -\sum_{i=1}^d p_i \log p_i,$$

the minimal entropy of any basis measurement of ρ , corresponding to the “best” basis, that is the eigenbasis of ρ .

Maassen-Uffink Uncertainty Principle

We've seen that for a basis $\mathbb{O} = \{|\phi_k\rangle\}$ we have

$$\log d \geq H(\mathbb{O}) \geq H(\rho),$$

but what about pairs of observables?

Let's take $\mathbb{O} = \{|\phi_k\rangle\}$ and also $\mathbb{O}' = \{|\xi_k\rangle\}$ s.t. $|\xi_k\rangle = \mathbf{U} |\phi_k\rangle$.

We now look for a bound on the sum of Shannon entropies

$$H(\mathbb{O}) + H(\mathbb{O}') \geq f(\mathbb{O}, \mathbb{O}'),$$

and, to do this, we will get a bound on the sum of Rényi entropies

$$H_\alpha(\mathbb{O}) + H_\beta(\mathbb{O}') \geq f(\alpha, \beta, \mathbb{O}, \mathbb{O}'),$$

for $\frac{1}{\alpha} + \frac{1}{\beta} = 2$, and take the limit $\alpha, \beta \rightarrow 1$.

Maassen-Uffink Uncertainty Principle

To start, consider a pure $|\psi\rangle$ and notice that

$$\sum_{k=1}^d |\phi_k\rangle \underbrace{\langle \phi_k | \psi \rangle}_{x_k} = |\psi\rangle = \sum_{k=1}^d |\xi_k\rangle \underbrace{\langle \xi_k | \psi \rangle}_{y_k},$$

where $y_k = \langle \xi_k | \psi \rangle = \langle \xi_k | \sum_{j=1}^n |\phi_j\rangle \langle \phi_j | \psi \rangle = \sum_{j=1}^n \mathbf{U}_{k,j} x_j$ so

$$H_\alpha(O) = \frac{1}{1-\alpha} \log \sum_{k=1}^d |x_k|^{2\alpha} = \frac{2\alpha}{1-\alpha} \log \|\mathbf{x}\|_{2\alpha},$$

$$H_\beta(O') = \frac{2\beta}{1-\beta} \log \|\mathbf{y}\|_{2\beta} = \frac{2\beta}{1-\beta} \log \|\mathbf{U}\mathbf{x}\|_{2\beta},$$

so we need to look for a relation between p -norms noting that the powers are harmonically conjugate i.e. $\frac{1}{2\alpha} + \frac{1}{2\beta} = 1$.

Riesz-Thorin Interpolation Theorem

Let $(\Omega_1, \Sigma_1, \mu_1)$ and $(\Omega_2, \Sigma_2, \mu_2)$ be σ -finite measure spaces.

Let $1 \leq p_0, p_1, q_0, q_1 \leq \infty$ and let $\mathbf{U}$ be an operator that maps $L^{p_0}(\mu_1) \mapsto L^{q_0}(\mu_2)$ and $L^{p_1}(\mu_1) \mapsto L^{q_1}(\mu_2)$. For any $0 < \theta < 1$, we denote $\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$ and $\frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}$. Then

$$\|\mathbf{U}\|_{p_\theta \rightarrow q_\theta} \leq \|\mathbf{U}\|_{p_0 \rightarrow q_0}^{1-\theta} \|\mathbf{U}\|_{p_1 \rightarrow q_1}^\theta \quad \text{i.e.}$$

$$\sup \frac{\|\mathbf{U}\mathbf{x}\|_{q_\theta}}{\|\mathbf{x}\|_{p_\theta}} \leq \left(\sup \frac{\|\mathbf{U}\mathbf{x}\|_{q_0}}{\|\mathbf{x}\|_{p_0}} \right)^{1-\theta} \left(\sup \frac{\|\mathbf{U}\mathbf{x}\|_{q_1}}{\|\mathbf{x}\|_{p_1}} \right)^\theta.$$

Taking $p_0 = q_0 = 2$, for unitary $\mathbf{U}$ we have $\|\mathbf{U}\mathbf{x}\|_2 = \|\mathbf{x}\|_2$.

Taking $p_1 = 1, q_1 = \infty$, $\|\mathbf{U}\mathbf{x}\|_\infty = \max_k |\mathbf{U}\mathbf{x}|_k \leq \max_{k,j} |U_{k,j}| \max_i |x_i|$.

We want $p_\theta = \frac{2}{1+\theta} = 2\alpha$ so $1 - \theta = 2 - \frac{1}{\alpha} = \frac{1}{\beta}$, $q_\theta = \frac{2}{1-\theta} = 2\beta$.

We conclude that $\|\mathbf{U}\mathbf{x}\|_{2\beta} \leq (\max_{k,j} |U_{k,j}|)^{1-1/\beta} \|\mathbf{x}\|_{2\alpha}$.

For convenience, we will denote $c = \max_{k,j} |U_{k,j}|$.

Maassen-Uffink Uncertainty Principle

We rewrite $\|\mathbf{U}\mathbf{x}\|_{2\beta} \leq c^{1-1/\beta} \|\mathbf{x}\|_{2\alpha}$ in terms of entropy as

$$\begin{aligned} \exp\left(\frac{1-\beta}{2\beta} H_{\beta}(O')\right) &\leq c^{1-1/\beta} \exp\left(\frac{1-\alpha}{2\alpha} H_{\alpha}(O)\right) \\ \Leftrightarrow \frac{1-\beta}{2\beta} H_{\beta}(O') &\leq (1-1/\beta) \log c + \frac{1-\alpha}{2\alpha} H_{\alpha}(O) \\ \Leftrightarrow \frac{1-\alpha}{2\alpha} H_{\alpha}(O) - \frac{1-\beta}{2\beta} H_{\beta}(O') &\geq -(1-1/\beta) \log c \\ \Leftrightarrow \left(\frac{1}{2} - \frac{1}{2\beta}\right) H_{\alpha}(O) + \left(\frac{1}{2} - \frac{1}{2\beta}\right) H_{\beta}(O') &\geq -2 \left(\frac{1}{2} - \frac{1}{2\beta}\right) \log c \\ \Leftrightarrow H_{\alpha}(O) + H_{\beta}(O') &\geq -2 \log c \end{aligned}$$

so we obtain the Maassen-Uffink Uncertainty Principle,

$$H(O) + H(O') \geq -\log \max_{k,j} |\langle \phi_k | \xi_j \rangle|^2.$$

which is valid for any $|\psi_i\rangle$ in the composition of ρ , so, taking the average, since H is concave, it is also valid for ρ itself.

Maassen-Uffink Uncertainty Principle

The Maassen-Uffink Uncertainty Principle states that for any pair of observables $\mathbf{O} = \sum_k \lambda_k |\phi_k\rangle \langle \phi_k|$ and $\mathbf{O}' = \sum_j \mu_j |\xi_j\rangle \langle \xi_j|$ with no duplicate eigenvalues, and for any state ρ we have

$$H(\mathbf{O}) + H(\mathbf{O}') \geq -\log \max_{k,j} \underbrace{|\langle \phi_k | \xi_j \rangle|^2}_{\text{overlap}}.$$

The lower bound is only zero if they share an eigenvector.

The lower bound is maximized when they are complementary.

Two observables are complementary if the maximal overlap is $1/d$, e.g. products of different Pauli matrices like $\mathbf{X} \otimes \mathbf{Z} \otimes \mathbf{Y}, \mathbf{Y} \otimes \mathbf{X} \otimes \mathbf{X}$.

This has a surprising number of applications.

Among them, security proofs for quantum key distribution.

The tighter the inequality, the more provably secure keyrate.

As better inequalities were developed, proofs also became easier.

Follow-up Reading

- ▶ J.Watrous – The theory of quantum information
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- ▶ P.Coles et al – Entropic uncertainty relations and their applications
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- ▶ H.Maassen,J.B.M.Uffink – Generalized entropic uncertainty relations
Physical review letters 60.12 (1988): 1103
- ▶ G.H.Hardy,J.E.Littlewood,G.Polya – Inequalities
Cambridge University Press (1934)
- ▶ M.Koashi – Simple security proof of quantum key distribution based on complementarity.
New Journal of Physics 11.4 (2009): 045018